

Let me update my memory first.Good, memory updated. Now let me rethink the signal categories from scratch — no paid API dependencies, global market lens, and with the depth needed to finalize this for v1 scoring.

Harvin Score: Final Signal Architecture (v1)

The Constraint-Aware Approach

Every signal below must pass three gates before it earns a place in the scoring system:

1. **Detectable without paid APIs** — scraping, public datasets, free tiers, DNS/HTTP inspection, or data already flowing through your existing pipeline
2. **Globally applicable** — works for a beauty brand in São Paulo, a fashion brand in Dubai, and a supplements brand in Jakarta. Not dependent on region-specific sources
3. **Maintainable by 1 data ops person** — if a signal requires daily manual verification across 50K accounts, it's out

EXTERNAL SIGNALS — Final v1 Set

Signal 1: Funding Events

What it captures: Capital injection that unlocks spending

Why it's the strongest signal: Across every geography, funded brands spend aggressively in the 90-day post-close window. This is universal — a Series A brand in Berlin behaves the same as one in Bangalore when it comes to tool/agency purchasing velocity.

Detection without paid APIs:

Source	Method	Coverage	Cost
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News scraping (RSS + HTML)	Monitor startup/business news sites per region. TechCrunch, EU-Startups, e27 (SEA), Wamda (MENA), Inc42 (India), Tech in Asia, The Ken, etc. Set up RSS feeds + keyword-filtered scraping for "raised," "funding," "Series," "seed round," "investment"	60-70% of funded D2C brands globally	Free
Twitter/X monitoring	Founders and investors frequently announce rounds on X before press coverage. Monitor accounts of known D2C investors and accelerators. Keyword search: "thrilled to announce," "excited to back," "pre-seed," "Series A"	Catches 15-20% that news misses	Free (X scraping or basic API free tier)
LinkedIn post monitoring	Similar founder/investor announcements. LinkedIn is particularly strong for European and SEA founders who post round announcements	Supplementary	Free (public post scraping)
Government/registry filings	Companies House (UK), MCA (India), SEC (US — for larger rounds), ACRA (Singapore). Funding rounds often appear in regulatory filings before news coverage	Catches stealth rounds	Free (public databases)
Accelerator/incubator batch announcements	YC, Techstars, 500 Global, Antler, Sequoia Surge, Entrepreneur First — all publish batch lists publicly	Very high precision	Free

Scoring tiers:

Sub-Signal	Points (out of 25 max)	Decay
Funding closed, 0–30 days ago	25	—
Funding closed, 31–60 days ago	20	Linear
Funding closed, 61–90 days ago	14	Linear
Funding closed, 91–180 days ago	7	Linear

Accelerator batch (current/recent)	15	90-day window
180+ days ago	0	Expired

Edge cases to handle:

- **Undisclosed amounts:** Still score high. The event happened; amount is secondary. Score as equivalent to the lowest band for their apparent stage.
- **Debt/venture debt rounds:** Score lower than equity rounds. Debt doesn't create the same spend pressure.
- **Multiple rounds in short succession (bridge → Series A):** Don't double-count. Take the most recent, highest-value round.

Signal 2: Growth Velocity

What it captures: Brand is scaling faster than its peers — meaning it's outgrowing its current tools and processes.

Detection without paid APIs:

Data Point	Free Source	Method	Refresh
Web traffic trend	Chrome CrUX dataset	Google's free, public dataset of real Chrome user traffic data. Covers millions of origins globally. You get traffic rank bands, not exact numbers — perfect for your MAU bands. Compare current period to previous to detect growth.	Monthly (CrUX updates monthly)
Web traffic rank	Tranco list	Free, daily-updated domain ranking list combining multiple sources. Track rank changes over time — a domain jumping from rank 500K to 100K is a clear growth signal.	Daily (but process monthly)
App growth signals	App Store / Play Store scraping	Scrape app listing pages for: rating count changes (more ratings = more downloads), version history (frequent updates = active development), "What's New" text, featured/editorial placements	Monthly

App store ranking changes	Same scraping	Track category rankings over time. A brand jumping from #200 to #50 in "Shopping" category = significant growth	Weekly
Social proof velocity	Instagram/Facebook public page scraping	Don't track absolute follower count (gameable). Track rate of engagement growth — comment counts, like ratios on recent posts. A brand whose engagement doubles in 3 months is growing fast.	Monthly
Website content velocity	Sitemap scraping	Parse sitemap.xml for new product pages, collection pages, blog posts. A brand adding 50 new product pages in a month is scaling their catalog.	Monthly
Google Trends relative interest	Google Trends (free, no API needed — scrape or use pytrends library)	Track brand name search interest over time. Rising search volume = growing consumer awareness. Works globally.	Monthly

Scoring tiers:

Sub-Signal	Points (out of 20 max)	Notes
MAU band upgrade (jumped a tier)	20	Strongest growth signal
Significant rank improvement on Tranco (>50% rank jump)	15	Validated by independent source
App rating count surge (>2x in 90 days)	12	Proxy for download growth
App category ranking jump (entered top 100)	10	Significant momentum
Google Trends breakout (>3x search interest in 90 days)	10	Consumer awareness growing
High content velocity (50+ new product pages in 30 days)	8	Catalog expansion
Steady / no measurable change	2	Baseline — still an active brand

Why I dropped social follower counts as a high-weight signal: Followers are bought, gamed, and inflated globally. A brand with 500K Instagram followers and 200 likes per post

is less valuable than one with 20K followers and 2,000 likes. Engagement rate matters more, but it's harder to track consistently across platforms and geographies. Keep it as a low-weight supplementary signal, not a core scoring input.

Signal 3: Operational Expansion

What it captures: The brand is expanding its physical, geographic, or channel presence — each expansion creates specific buying needs.

Detection without paid APIs:

Signal	Detection Method	Cost	Global Applicability
New offline store openings	Scrape the brand's own "Store Locator" page. Most omnichannel brands have one at /stores, /locations, /store-locator. Parse it monthly, compare to previous month. Delta = new stores.	Free	Works globally — store locator pages follow similar patterns everywhere
Marketplace expansion	Search for the brand on major marketplaces per region. Amazon (global), Shopee/Lazada (SEA), Noon (MENA), Flipkart/Myntra (India), Zalando (EU). If a brand appears on a marketplace it wasn't on last month, that's a signal.	Free	Adapt marketplace list per region
Quick commerce entry	Same approach — search Blinkit/Zepto (India), GrabMart (SEA), Getir (EU/MENA), Gopuff (US). Quick commerce entry is a strong signal that the brand is scaling distribution aggressively.	Free	Region-specific marketplaces
D2C website launch (was marketplace-only)	Monitor your "marketplace-only" accounts for new domain activity. If an account you tagged as "Only Marketplace" suddenly has a working Shopify/custom site, that's a massive signal.	Free	Universal

International expansion	Detect new language/currency options on website. Scrape for hreflang tags, currency selectors, country dropdowns. If a brand adds Arabic or Bahasa or German to their site, they're expanding. Also detect new country-specific domains (brand.co.uk, brand.de).	Free	Universal — hreflang and currency detection works everywhere
New warehouse/fulfillment signals	Job postings in new cities for operations/logistics roles. If a Mumbai brand posts warehouse jobs in Dubai, they're expanding to MENA.	Free (job board scraping)	Universal

Scoring tiers:

Sub-Signal	Points (out of 20 max)	Notes
D2C website launch (was marketplace-only)	20	Highest single signal — they need EVERYTHING
First offline store (online-only → omnichannel)	18	Business model transition = new tool needs
New store openings (existing omnichannel, 3+ stores)	12	Continued expansion
New marketplace entry (first time on a platform)	10	Channel diversification
Quick commerce entry	10	Aggressive distribution play
International expansion detected	15	Signals ambition + budget
Single new store or minor expansion	5	Activity, but lower urgency

Signal 4: Hiring Intent

What it captures: Hiring patterns predict purchasing decisions 30–90 days in advance. A new VP Marketing will evaluate and buy tools within their first quarter.

Detection without paid APIs:

Source	Method	Coverage	Cost
LinkedIn Jobs (public scraping)	LinkedIn public job listings are scrapable without API. Search by company name, filter by role keywords.	High coverage globally — LinkedIn is the dominant professional platform everywhere except China	Free
Google Jobs aggregation	Google Jobs aggregates from multiple sources (Indeed, Glassdoor, company career pages). Scrape Google Jobs search results for your tracked brands.	Very high coverage	Free
Company career pages	Many D2C brands have /careers or /jobs pages. Scrape these directly.	Medium (not all brands have career pages)	Free
Indeed / Glassdoor / regional job boards	Naukri (India), JobStreet (SEA), Bayt (MENA), StepStone (EU). Scrape for tracked brand names.	Supplementary	Free

Role classification system (build this as a simple rules-based mapper):

Role Tier	Example Titles	What It Signals	Score Multiplier
Tier 1: Executive	VP Marketing, CMO, CTO, VP Growth, VP Ecommerce, COO	Major strategic shifts incoming. New executive = new vendor evaluation cycle.	1.0x (full weight)
Tier 2: Senior Leader	Head of Growth, Head of Digital, Director of Marketing, Head of D2C, Head of Data	Department-level changes. Budget authority. Tool/agency decision-maker.	0.75x
Tier 3: Mid-Level	Marketing Manager, Ecommerce Manager, Performance Marketing Lead, CRM Manager	Operational hiring. Indicates growth but less likely to trigger vendor changes immediately.	0.4x
Tier 4: Individual Contributor	Social Media Executive, Content Writer, Graphic Designer, Customer Support	General growth signal but no purchasing authority.	0.15x

Scoring tiers:

Sub-Signal	Points (out of 15 max)	Notes
Tier 1 executive hire in marketing/growth	15	Strongest hiring signal
Tier 1 executive hire in tech/data	12	Tech stack review likely
Tier 2 leader hire	10	Department-level intent
Bulk hiring (10+ open roles simultaneously)	10	Scaling signal regardless of role types
Team size threshold crossed (e.g., 50→100)	8	Organizational milestone
Tier 3 mid-level hire	5	Moderate signal
Tier 4 IC hires only	2	Low but non-zero signal

Global nuance: Job titles vary by region. "Chief Growth Officer" is common in the US; "Head of Performance" is more common in India and SEA. Your role-tier mapping needs regional aliases. Build this as a YAML config (same pattern as tech stack detection) so data ops can add title variations without code changes.

Signal 5: Tech Stack Changes

What it captures: Changes in a brand's technology infrastructure — additions, removals, and migrations that reveal intent and create sales opportunities.

Detection: This is entirely powered by your existing tech stack detection engine (the 101-tool YAML config system). The signal comes from **delta analysis** — comparing the current month's scan to the previous month's scan.

Delta types and what they mean:

Delta Type	What Happened	What It Means	Points (out of 20 max)
Engagement tool removed	Brand was using MoEngage last month, not detected this month	Either churned the tool or in active evaluation. A competitor's sales rep should call immediately. Also signals to the removed tool's vendor that churn happened.	20

Greenfield + growing	No engagement platform detected AND traffic is growing	Dream lead for engagement SaaS. Zero competitive friction + proven need.	20
Ecommerce platform migration	Changed from WooCommerce to Shopify, or Shopify to custom	Active infrastructure investment. Brand is spending on fundamentals. Open to adjacent tool evaluation.	15
Payment gateway change/addition	Added Stripe alongside Razorpay, or switched entirely	International expansion (Stripe for global, Razorpay for India). Signals payment optimization focus.	10
Analytics tool addition	Added Mixpanel, Amplitude, or Hotjar	Investing in data-driven decision making. Likely receptive to other optimization tools.	8
Support tool change	Switched from Tawk.to (free) to Zendesk or Intercom	Upgrading customer support = brand is maturing and investing in CX. Budget is growing.	8
New tool in any category	Any new tool detected that wasn't there before	General tech investment signal	5
No change detected	Same stack as last month	Stable. No urgency signal.	0

🚩 **Critical implementation detail:** You need to store historical tech stack snapshots to compute deltas. In your database schema, don't just store the current tech stack — store every scan result with a timestamp. The `tech_scan_log` table already tracks `last_scanned`, but you need a `tech_stack_history` table:

tech_stack_history:

- account_id
- category
- tool_name
- first_detected_at
- last_detected_at
- status (active / removed)
- scan_month

This gives you the longitudinal data that becomes your moat. By month 12, you'll have 12 snapshots per account — a year of tech stack evolution data that no competitor can replicate without starting their own detection engine and waiting 12 months.

Signal 6: Brand Maturity & Digital Sophistication

What it captures: How "ready to buy" a brand is based on its digital maturity stage. This is a structural signal, not an event signal — it changes slowly but frames all other signals.

Detection without paid APIs:

Indicator	Detection Method	What It Tells You	Cost
Website sophistication	Automated page quality scoring: Does the site have a proper favicon? Custom domain (not myshopify.com)? SSL certificate? Multiple pages? Blog? Structured data (schema.org markup)? Mobile responsiveness? Each is a simple check during your regular scraping.	A brand with a polished, multi-page site with structured data is more mature (and has more budget) than one running a basic Shopify template on a myshopify.com subdomain.	Free
Custom domain vs. platform subdomain	Check if domain is brand.com vs. brand.myshopify.com or brand.mydukaan.io	Custom domain = more invested in their D2C presence	Free (DNS check)
Multi-language / multi-currency support	Detect hreflang tags, currency selectors, country pickers	Serving multiple markets = more complex operations = bigger budgets	Free
Active blog / content marketing	Check for /blog, /journal, /stories pages with recent content (last 30 days)	Content investment signals marketing maturity and budget	Free
Structured data / SEO investment	Check for JSON-LD schema markup, Open Graph tags, proper meta descriptions	SEO investment signals digital marketing sophistication	Free
Email marketing visible	Detect email capture popups, newsletter signup forms, exit-intent modals	Active email strategy = likely using or ready to use engagement tools	Free
Reviews/UGC integration	Detect review widgets (Judge.me, Yotpo, Stamped, Loox)	Social proof investment = conversion optimization mindset	Free (part of tech stack detection)

This isn't scored as individual events — it's a baseline "readiness" multiplier:

Maturity Level	Indicators	Score Multiplier
High maturity	Custom domain + multi-language + active blog + structured data + reviews	1.2x (boosts all other signals by 20%)
Medium maturity	Custom domain + some optimization signals	1.0x (no adjustment)
Low maturity	Platform subdomain, minimal optimization, no blog	0.7x (dampens other signals — brand may not have budget yet)

Why a multiplier instead of additive points: A newly funded brand with a polished, multi-market website (high maturity) is a much hotter prospect than a newly funded brand running a bare myshopify.com store. The funding signal is the same, but the readiness to act is different. The maturity multiplier captures this.

INTERNAL SIGNALS — Final v1 Set

What to Track From Day One

You won't use these in the score formula at launch (not enough data). But if you don't instrument them from day one, you'll lose months of training data. **Build the tracking, store the events, deploy the model later.**

Signal	Event to Log	Schema	Purpose
Account detail page view	<code>user_id, account_id, timestamp, source (explorer/feed/watchlist/extension/search), session_id</code>	High-volume, low-intent signal. Establishes baseline interest.	Train similarity model ("users who view X also view Y")
Watchlist addition	<code>user_id, account_id, watchlist_id, timestamp, method (manual/filter/csv)</code>	Core intent signal. User explicitly decided this account matters.	Train scoring model — watchlisted accounts are "positive" labels

Watchlist removal	<code>user_id, account_id, watchlist_id, timestamp, reason (optional dropdown: "not relevant", "already contacted", "closed deal", "lost deal", "other")</code>	Negative signal + outcome data if reason provided	Train what makes an account NOT valuable
CRM push	<code>user_id, account_id, crm_type (hubspot/salesforce), timestamp</code>	Strongest intent signal. User is taking action.	Highest-weight positive label for scoring
CSV export	<code>user_id, account_ids[], timestamp, source (explorer/watchlist)</code>	Action-oriented signal. User is moving data to their workflow.	Positive label, slightly lower weight than CRM push
Alert click (Slack/email)	<code>user_id, account_id, alert_type, signal_type, timestamp</code>	Signal was relevant enough to act on	Train alert ranking model
Alert dismiss	<code>user_id, account_id, alert_type, signal_type, timestamp</code>	Signal was NOT relevant	Negative label for alert ranking
Search query	<code>user_id, query_text, timestamp, results_count</code>	Reveals what users are looking for	Train search ranking + identify gaps
Filter combination used	<code>user_id, filters_applied (JSON), results_count, timestamp</code>	Reveals how users segment the market	Identify high-value micro-segments
Chrome Extension click	<code>user_id, domain, timestamp, action (view/watchlist/crm)</code>	Real-world	Context-rich intent signal

		research behavior	
Session duration on account page	<code>user_id, account_id, duration_seconds, timestamp</code>	Dwell time = depth of interest	Supplement ary intent signal
Outcome tag (manual)	<code>user_id, account_id, status (contacted/in_pipeline/won/lost/not_ relevant), timestamp, notes (optional)</code>	THE PMF signal. Ties Harvin intelligenc e to real revenue outcomes	Defines what a "good" account actually looks like

The Manual Outcome Tagging System (Ship in v1)

This is simple to build but strategically critical. On every watchlisted account, show a small status indicator:

States:

-  **New** (default — just added to watchlist)
-  **Contacted** — user reached out
-  **In Pipeline** — active opportunity
-  **Won** — deal closed
-  **Lost** — deal didn't work out
-  **Not Relevant** — removed from consideration

When a user marks "Won" or "Lost," optionally prompt for a one-line note. Don't force it, but capture it if offered.

Why this matters more than CRM sync:

- It's YOUR data in YOUR platform (no API dependency, no sync complexity)
- It captures outcomes for users who don't even use a CRM (many agency founders don't)
- It creates investment in the platform (updating statuses = building workflow habits = switching cost)
- It gives you ground truth labels for ML scoring: "Won" accounts define what a good recommendation looks like

Precedent: LinkedIn Sales Navigator's "Lead Status" tags. Simple, manual, no CRM sync required. Yet it's one of the stickiest features because users build their pipeline view inside the product.

FINAL V1 SCORING FORMULA

Architecture: Component Scores → Weighted Sum → Maturity Multiplier → Final Score

Raw Score = (Capital × 0.25) + (Growth × 0.20) + (Expansion × 0.20)
+ (Hiring × 0.15) + (TechStack × 0.20)

Final Score = Raw Score × Maturity Multiplier

Capped at 100.

Why These Weights

Component	Weight	Rationale
Capital (Funding)	25%	Most reliable buying signal across all personas and geographies. Universal.
Growth Velocity	20%	Brands outgrowing their tools create organic demand. Strong for SaaS sales, strong for agencies.
Operational Expansion	20%	Physical/channel expansion creates specific, actionable buying needs. Strong for agencies especially.
Hiring Intent	15%	Leading indicator but noisier than the others. Title parsing has error rates. Lower weight until accuracy improves.
Tech Stack Changes	20%	Harvin-unique signal. No competitor tracks this. Weighted high to differentiate scoring.

Score Display — What the User Sees

On every account card and detail page:

Score: 87 🔥 Hot

WHY:

- 💰 Raised \$5M Series A (18 days ago)
- 📈 Traffic surged 3x this quarter
- 🏪 Opened 4 new stores
- 👤 Hiring Head of Growth
- 🔧 No engagement platform detected

The "WHY" breakdown is non-negotiable. A naked number (87) is meaningless. The explanation is what makes a user trust the score and act on it. It's also what makes Harvin feel like intelligence, not just data.

Decay Mechanics

All event-based signals (funding, hiring, store openings) decay linearly:

Age of Signal	Score Retention
0–30 days	100%
31–60 days	75%
61–90 days	50%
91–120 days	25%
121+ days	0% (signal expires)

Structural signals (growth velocity, maturity, tech stack state) don't decay — they refresh when re-measured (monthly).

Score recalculation schedule: Nightly batch job. Recalculate scores for any account that had a new signal detected, a tech stack scan update, or a traffic data refresh. Don't recalculate all 50K every night — only accounts with changed inputs. Should affect 2,000–5,000 accounts per night.

Signals Explicitly Cut From v1

Signal	Why Cut	When to Add
Social media follower counts	Gameable, noisy, poor proxy for buying intent globally	v2, only as engagement rate (not absolute count)
Revenue estimates	Impossible to get accurately for private companies. Wrong revenue data destroys trust instantly.	Maybe never. Revenue estimation for private D2C brands is a losing game.
Ad spend signals (running Google Ads, Meta Ads)	Extremely volatile (campaigns start and stop weekly), hard to detect without paid tools (SEMrush/SpyFu), and a maintenance nightmare at 50K accounts	v3, and only if a free detection method emerges

PR/media mentions volume	Noisy, hard to distinguish paid PR from organic, and the signal is ambiguous (is press coverage good or bad?)	v2, as a supplementary signal only
Customer review sentiment	Valuable in theory, but sentiment analysis across multiple languages and platforms is a Tier 3 ML problem. Don't try this with a rules engine.	v3, with proper NLP investment
Competitor's customer list changes	"Which brands switched FROM MoEngage TO CleverTap" — incredibly valuable but requires two sequential scans to be accurate, and attribution is uncertain	v1.5, once you have 3+ monthly snapshots

The 30-Second Summary

Six external signal categories power the v1 Harvin Score, all detectable without paid APIs:

1. **Capital injection** — funding events, accelerator participation
2. **Growth velocity** — traffic surges, app growth, search interest
3. **Operational expansion** — stores, marketplaces, new markets, D2C site launches
4. **Hiring intent** — senior role postings with seniority-tiered weighting
5. **Tech stack changes** — tool removals, greenfield detection, migrations
6. **Brand maturity** — digital sophistication as a readiness multiplier

Internal signals are instrumented from day one but don't enter the score formula until you have enough behavioral data (target: 3 months post-beta, 50+ active users).

The outcome tagging system ships in v1 as a simple manual status tracker on watchlisted accounts — this is how you measure PMF without CRM integration complexity.

Want me to dive into the specific data pipeline architecture for any of these signals, or move to designing how the score and its breakdown surfaces across the UI — Market Intelligence feed, Account Explorer cards, Account Detail page, and alerts?

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